

A	B	C	D	E	F
G	H	I	J	K	L
M	N	O	P	Q	R
S	T	U	V	W	X
Y	Z	1	2	3	4
5	6	7	8	9	

P300 speller

decode



Neural decoder

post-edit



Language model

adjudicate



Physician panel (2)

INTENDED MESSAGE

NOISY DECODER TEXT

MODEL OUTPUT

CONCORDANT PHYSICIAN LABEL

FAITHFUL

meaning preserved

I WANT TO CONTINUE THE FEEDING TUBE

gemma4:31b · CRIT · target CER 20%

I WAN TO CONTIGUE THE IFN4ICA TUBE

20% realized CER

“I WANT TO CONTINUE THE FEEDING TUBE”

Faithful

WHAT ARE YOU TALKING ABOUT?

qwen2.5:7b · AUTH · target CER 30%

WHATRE1YOZ TAW4ING5ABOZT?

30% realized CER

“WHAT ARE YOU TALKING ABOUT?”

Faithful

DEGRADED

output visibly breaks down

WOULD YOU LIKE TO GO WITH ME?

phi4:mini · CTRL · target CER 20%

I0U3D YOU LTKE TO G0C W7TH1ME?

21% realized CER

“I Owe You To Get With Me?”

Degraded

PLEASE LET ME FINISH

mistral-small:24b · CTRL · target CER 30%

PME8SE LEX MH BRNISH

30% realized CER

“PEACE LX M BRITISH”

Degraded

DRIFT

fluent, but a different message

I'M DOING THE IMPOSSIBLE

mistral-small:24b · AUTH · target CER 0%

I'M DOING THE IMPOSSIBLE

0% realized CER

“I'M DOING THE POSSIBLE”

Drift

WHERE?

gemma4:31b · CTRL · target CER 40%

8BKRJ?

67% realized CER

“Hello?”

Drift